laced with pre-Copernican thinking, a more sophisticated interpretation of Hubble's discovery is called for, one in which our location does not occupy some special place in the cosmic order.

General relativity provides such an interpretation. With general relativity, Einstein found that space and time are flexible, not fixed, rubbery, not rigid; and he provided equations that tell us precisely how space and time respond to the presence of matter and energy. In the 1920s, the Russian mathematician and meteorologist Alexander Friedmann and the Belgian priest and astronomer Georges Lemaître independently analyzed Einstein's equations as they apply to the entire universe, and the two found something striking. Just as the gravitational pull of the earth implies that a baseball popped high above the catcher must either be heading farther upward or must be heading downward but certainly cannot be staying put (except for the single moment when it reaches its highest point), Friedmann and Lemaître realized that the gravitational pull of the matter and radiation spread throughout the entire cosmos implies that the fabric of space must either be stretching or contracting, but that it could not be staying fixed in size. In fact, this is one of the rare examples in which the metaphor not only captures the essence of the physics but also its mathematical content since, it turns out, the equations governing the baseball's height above the ground are nearly identical to Einstein's equations governing the size of the universe.[6]

The flexibility of space in general relativity provides a profound way to interpret Hubble's discovery. Rather than explaining the outward motion of galaxies by a cosmic version of the factory explosion, general relativity says that for billions of years space has been stretching. And as it has swelled, space has dragged the galaxies away from each other much as the black specks in a poppy seed muffin are dragged apart as the dough rises in baking. Thus, the origin of the outward motion is *not* an explosion that took place within space. Instead, the outward motion arises from the relentless outward swelling of space itself.

To grasp this key idea more fully, think also of the superbly useful balloon model of the expanding universe that physicists often invoke (an analogy that can be traced at least as far back as a playful cartoon, which you can see in the endnotes, that appeared in a Dutch newspaper in 1930